Class: SE **Subject: COA LAB** **Sem-III**

Experiment No 3

Course Outcome: CO2 **Blooms Level: L3**

Aim: Implementation of number conversion (HEX to BCD, ASCII to BCD) using MASM.

Abstract:

1)Write an assembly program that takes a single-byte hexadecimal value, for example, 3Ch. The main goal is to convert this hex number into its Binary-Coded Decimal (BCD) form by separating it into its tens and units decimal digits (like 6 and 0 for 3Ch). After getting these BCD digits, the program must change each digit into its matching ASCII character. Finally, the program needs to print these ASCII characters one after the other to the console. The output for 3Ch must be exactly '60', showing the hexadecimal value as a clear decimal string.

2)Create an 8086 assembly language program that allows the user to input a two-digit decimal number ranging from 00 to 99 using keyboard input. The program should read the individual digits (tens and units), convert them from ASCII to numeric values, and combine them into a single packed Binary-Coded Decimal (BCD) byte. The packed BCD should represent the tens digit in the upper nibble and the units digit in the lower nibble. After packing the BCD value, the program should display the result in hexadecimal format on the screen.

Sample Input:

Case 1:HEX TO BCD:3CH
Case 2:ASCII TO BCD:Input through prompt.

Theory: Computers process information in binary, but humans primarily interact with decimal numbers and characters. This necessitates various number systems and conversion techniques.

Binary Coded Decimal (BCD) is a method to represent decimal numbers where each decimal digit is encoded into its own 4-bit binary value (a nibble). Packed BCD stores two such decimal digits within a single 8-bit byte. BCD is particularly useful for applications requiring exact decimal arithmetic, like financial calculations, or when directly interfacing with numerical displays.

Hexadecimal (base-16) is a fundamental number system in computing, using digits 0-9 and letters A-F. Its efficiency comes from each hexadecimal digit directly representing four binary bits, making it a compact shorthand for binary data, memory addresses, and data values. Converting between hexadecimal and decimal is a common task for human readability and system understanding.

Beyond numbers, ASCII (American Standard Code for Information Interchange) is a character encoding standard that assigns unique numerical values (typically 7-bit, extended to 8-bit for common use) to letters, numbers, punctuation, and control characters. When users input characters like '5', they are received as their ASCII code, which then needs to be converted to its numerical value for arithmetic operations, or vice-versa for display. This process of converting between character codes and numerical values is essential for virtually all user interaction with computer systems.

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Program:

1) HEX TO BCD

```
DATA SEGMENT
X  DB 0B2H
   DB 100 DUP (00)
DATA ENDS
CODE SEGMENT
ASSUME CS: CODE, DS: DATA
START:  MOV AX, DATA
        MOV DS, AX
        MOV AX, 0000H
        MOV BX, 0000H
        MOV AL, X
        MOV CL, 0AH
REPEAT: DIV CL
        MOV [BX+8], AH
        INC BX
        MOV AH, 00H
        CMP AL, CL
        JNC REPEAT
```

```
        MOV [BX+8], AL
        INT 3H
CODE ENDS
END START
```

2) ASCII TO BCD

```
DATA SEGMENT
ASCII DB 31H, 32H, 33H, 34H, 35H, 36H, 37H, 38H, 39H, 30H
        DB 100 DUP(00H)
DATA ENDS
CODE SEGMENT
ASSUME CS: CODE, DS: DATA
START:  MOV AX, DATA
        MOV DS, AX
        MOV CX, 10
        MOV BX, 0000H
BACK:   MOV AL, [BX]
        AND AL, 0FH
        MOV [BX+16], AL
        INC BX
```

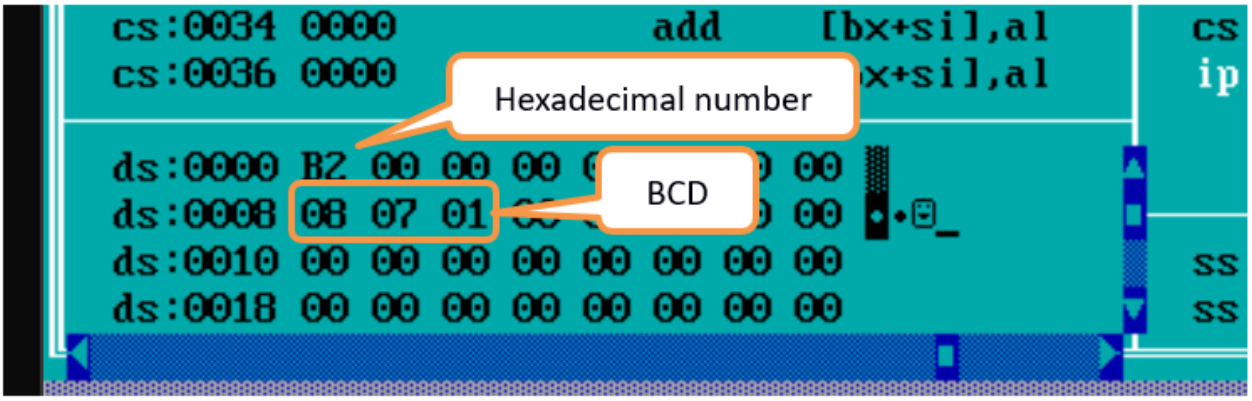
```
        LOOP BACK

        INT 3H

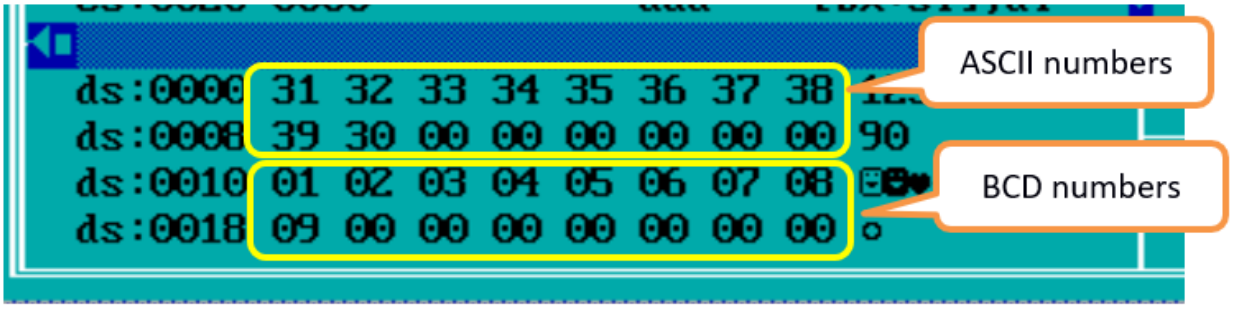
CODE ENDS

END START
```

Output: Output window: (a) Hexadecimal to BCD conversion



Output window: (b) ASCII to BCD conversion



Conclusion:
The experiment on "Implementation of Number Conversion (HEX to BCD, ASCII to BCD) using MASM" was successfully carried out to demonstrate how numeric data is converted into

human-readable formats at the assembly language level using the 8086 microprocessor. This conversion is essential when displaying results on the screen or interacting with user input, as microprocessors operate primarily in binary or hexadecimal formats, while users typically work with decimal values.

The first program converts a hard-coded hexadecimal number (3Ch) into its equivalent decimal representation and displays it as two characters (tens and units). The logic uses the DIV instruction to divide the value by 10, storing the quotient and remainder as decimal digits. These are then converted to ASCII characters and printed using DOS interrupts. The program efficiently demonstrates how division can be used to extract individual decimal digits from a hexadecimal value.

The second assembly program successfully converts a decimal number into its BCD (Binary-Coded Decimal) format. It achieves this by reading the tens and units digits individually as ASCII characters, converting them to numerical values, and then combining them into a single byte where the higher nibble holds the tens digit and the lower nibble holds the units digit. This packed representation is then displayed on the screen using a custom routine that prints hexadecimal values. The program effectively demonstrates basic I/O handling, ASCII to numeric conversion, and bitwise manipulation in 8086 assembly.

Exercise 1:

Aim: To develop an assembly language program for converting a BCD number into its ASCII equivalent.

Abstract:
Design and implement an assembly language program using MASM that converts a BCD number into its corresponding ASCII representation. The program should accept a single byte where the upper nibble represents the tens digit and the lower nibble represents the units digit (valid BCD values: 00H to 99H). It should then extract both nibbles, convert each to its ASCII equivalent, and display the result on the screen.